Evaluating a Bayesian Network for Crop Recommendation: Predictive Performance, Environmental Sensitivity, and Uncertainty

Bhadresh RP
CB.SC.U4AIE24208
Department of Artificial Intelligence
Amrita Vishwa Vidyapeetham
Coimbatore, Tamil Nadu, India

Ch. Priya Madhav
CB.SC.U4AIE24211
Department of Artificial Intelligence
Amrita Vishwa Vidyapeetham
Coimbatore, Tamil Nadu, India

B. Cibi Kumar
CB.SC.U4AIE24212
Department of Artificial Intelligence
Amrita Vishwa Vidyapeetham
Coimbatore, Tamil Nadu, India

G.G. Rishi Kanth
CB.SC.U4AIE24218
Department of Artificial Intelligence
Amrita Vishwa Vidyapeetham
Coimbatore, Tamil Nadu, India

***Abstract*—Crop recommendation models use soil and environmental measurements to predict suitable crop classes. In addition to predictive accuracy, agricultural decision support can benefit from probability estimates that express uncertainty and allow the effect of changing conditions to be inspected. This paper evaluates a Bayesian Network (BN) for crop recommendation using nitrogen, phosphorus, potassium, temperature, humidity, soil pH, and rainfall. Numerical features are discretized into three quantile-based categories using boundaries derived from the training partition. A network structure is learned using hill-climbing search with the Bayesian Dirichlet equivalent uniform score, and conditional probability distributions are estimated using Bayesian parameter estimation. Variable elimination is used to infer posterior probabilities over 22 crop classes. On a stratified 80:20 holdout split of a 2,200-sample dataset, the BN achieves 75.68% accuracy and a macro F1-score of 0.73. On the same holdout split, conventional classifiers achieve higher predictive accuracy, with Random Forest, Extra Trees, and Gaussian Naive Bayes each reaching 99.55%. Scenario analysis with the BN shows that the predicted crop distribution changes under different rainfall evidence; the normalized entropy also varies across these scenarios. The results indicate that the current discretized BN is a useful probabilistic baseline but does not outperform standard classifiers on this dataset. Further work is needed to assess probability calibration, structural stability, and performance on independent agricultural data.**

***Index Terms*—Bayesian Network, Crop Recommendation, Probabilistic Inference, Classification, Sensitivity Analysis, Predictive Uncertainty**

I. Introduction

Crop selection depends on interacting soil and environmental conditions, including nutrient levels, temperature, humidity, soil pH, and rainfall. Data-driven crop recommendation has therefore been studied using conventional machine-learning classifiers and decision support systems [2]–[4].

Many crop recommendation systems report a single predicted class. A probabilistic model can additionally represent uncertainty over candidate classes and provide a framework for examining how the prediction changes when environmental evidence changes. Bayesian Networks (BNs) represent a joint probability distribution through a directed acyclic graph and conditional probability distributions. A BN can therefore support inference over crop classes while explicitly representing dependencies among variables.

However, probabilistic representation alone does not guarantee better predictive performance or well-calibrated confidence. Prior work has investigated Bayesian Networks for crop recommendation [1], while recent work has studied uncertainty-aware crop recommendation using machine learning [5]. Consequently, an evaluation of a BN should report both its predictive performance and the behavior of its posterior probabilities, and should compare it with conventional baselines.

This study evaluates an existing BN implementation using seven environmental features and 22 crop classes. Its objectives are:

- to describe the preprocessing, structure-learning, parameter-learning, and inference pipeline;
- to compare the BN's holdout performance with conventional classifiers under the same train-test split;
- to examine the effect of selected environmental evidence on posterior crop probabilities; and
- to summarize posterior uncertainty using normalized Shannon entropy.

The results are an empirical baseline. The study does not claim that the BN is more accurate than conventional classifiers, nor that the scenario results establish agronomic causality.

II. Related Work

Thilakarathne *et al.* presented a cloud-enabled crop recommendation platform for precision farming and evaluated multiple machine-learning approaches on a crop recommendation

TABLE I
DATASET SUMMARY

Property	Value
Total samples	2,200
Input features	7
Crop classes	22
Samples per class	100
Missing values	0
Exact duplicate rows	0
Training samples	1,760
Testing samples	440

dataset [2]. Prity *et al.* compared nine classification methods for crop recommendation and reported Random Forest as their strongest model in their experimental setting [3]. These studies illustrate the established role of conventional classifiers in crop recommendation.

Bayesian network classifiers provide a principled framework for probabilistic classification [9]. Bayesian probabilistic modeling has also been explored for agricultural decision support. Sharmin and Kiekintveld described a Bayesian Network-based smart crop recommendation system that uses probabilistic reasoning to consider site-specific parameters and their impacts [1]. This prior work means that using a BN for crop recommendation is not, by itself, a novel contribution.

Interpretability and uncertainty are additional themes in recent literature. Shastri *et al.* studied supervised crop recommendation with explainable artificial intelligence [4]. Alam *et al.* proposed a machine-learning framework with entropy-based uncertainty quantification for crop recommendation on a 2,200-sample dataset with 22 classes [5]. Thus, entropy-based uncertainty analysis on this dataset is also not a new contribution on its own.

The present work should therefore be understood as a reproducible empirical evaluation of a discretized BN, its sensitivity to selected evidence, and its comparison with conventional classifiers. A stronger novelty claim would require further experiments, such as probability calibration analysis, robustness across multiple splits, or independent agricultural validation.

III. Dataset and Preprocessing

A. Dataset Description

The supplied CSV contains 2,200 samples and eight columns: seven input features and one target label. The features are nitrogen (`n`), phosphorus (`p`), potassium (`k`), temperature, humidity, soil pH, and rainfall. The target contains 22 crop classes, with 100 records per class. The dataset structure matches the publicly described crop recommendation dataset available on Kaggle [14]. The provenance and original data-collection process should be checked before submission.

The supplied file has no missing values and no exact duplicate rows. The seven input features are retained without feature scaling for the BN, while selected conventional baselines use standardization where appropriate.

B. Train-Test Split

A stratified 80:20 train-test split is created with `random_state=42`. The training set contains 1,760 samples and the holdout test set contains 440 samples. The same split is used for the BN and conventional classifiers to make the holdout comparison consistent.

C. Feature Discretization

The BN implementation uses categorical states. Each numerical feature is discretized into Low, Medium, and High categories using quantile boundaries computed from the training data only. The boundaries are then applied to both training and test data.

For a numerical feature X, the discretization is

$$D(X) = \begin{cases} \text{Low}, & X \le b_1, \\ \text{Medium}, & b_1 < X \le b_2, \\ \text{High}, & X > b_2, \end{cases} \tag{1}$$

where b_1 and b_2 are the training-derived quantile boundaries. Discretization simplifies categorical probability estimation but may discard information about differences between numerical values.

IV. Bayesian Network Methodology

A. Network Representation

A BN consists of a directed acyclic graph $G = (V, E)$ and a conditional probability distribution for each node [8], [10]. The graph contains the seven environmental variables and the crop label. The joint distribution factorizes as

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \mathrm{Pa}(X_i)), \tag{2}$$

where $\mathrm{Pa}(X_i)$ denotes the parents of node X_i.

For environmental evidence E, inference estimates the posterior probability of each crop class C:

$$P(C \mid E). \tag{3}$$

The top-ranked recommendation is

$$\hat{C} = \arg\max_{c \in \mathcal{C}} P(C = c \mid E), \tag{4}$$

where $\mathcal{C}$ is the set of crop classes.

B. Structure and Parameter Learning

The network structure is learned using hill-climbing search with the Bayesian Dirichlet equivalent uniform (BDeu) score and an equivalent sample size of 10. Edges from the crop label to feature nodes are forbidden during structure learning. The learned graph contains 21 directed edges in the recorded experiment.

Conditional probability distributions are estimated using Bayesian parameter estimation with Dirichlet pseudo-counts of 0.1. The resulting network contains eight nodes and eight conditional probability distributions and passes the model-consistency check.

C. Probabilistic Inference

Variable elimination is used to compute the posterior distribution over crop labels for each test instance. The seven discretized feature states form the evidence. The class with the highest posterior probability is selected as the predicted crop.

V. Experimental Setup

A. Evaluation Metrics

Classification performance is measured using accuracy and macro F1-score. Accuracy is

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}. \tag{5}$$

For a class, the F1-score is

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{6}$$

Macro F1 averages class-wise F1 scores without weighting by class support.

Brier score and log loss are useful metrics for evaluating predicted class probabilities, because probability quality and classification accuracy are distinct aspects of performance [11]–[13]. The Brier score is the mean, across samples, of the sum of squared differences between the predicted class-probability vector and the one-hot target vector. Lower Brier score and lower log loss indicate better probabilistic predictions. Neither metric is reported in the current results tables: the saved BN evaluation did not include its full holdout posterior vectors, and the baseline comparison was limited to accuracy and F1-scores. Consequently, probability calibration and proper-scoring-rule performance cannot be compared in this study.

B. Baseline Models

The conventional baselines are Random Forest, Gaussian Naive Bayes, Extra Trees, Decision Tree, RBF Support Vector Machine, K-Nearest Neighbors, Logistic Regression, and Gradient Boosting. The classifiers use scikit-learn default settings except for fixed random seeds and a higher iteration limit for Logistic Regression. Standardization is applied within pipelines for SVM, KNN, and Logistic Regression. The baseline metrics are calculated on the same 440-sample holdout set as the BN. Before interpreting the very high baseline scores, the split and feature construction should be checked to ensure that the target label or any target-derived information is not included among the input features. The reported comparison uses the seven environmental features only; the large gap should not be attributed to discretization alone without a controlled ablation study.

A separate five-fold stratified cross-validation is reported for the conventional baselines using the training partition. These values are fold-based estimates and are distinct from the holdout results; they are not a replacement for independent external validation.

C. Sensitivity and Uncertainty

Sensitivity analysis changes one categorical feature at a time while holding the other baseline evidence states fixed. The total variation distance between two posterior distributions P and Q is

$$D_{\text{TV}}(P, Q) = \frac{1}{2}\sum_{c \in \mathcal{C}} |P(c) - Q(c)|. \tag{7}$$

A larger value indicates a larger change in the posterior distribution for the selected scenario.

Shannon entropy is calculated as

$$H(P) = -\sum_{c \in \mathcal{C}} P(c) \log P(c), \tag{8}$$

and normalized entropy for 22 classes is

$$H_{\text{norm}}(P) = \frac{H(P)}{\log 22}. \tag{9}$$

Normalized entropy lies between zero and one. It measures how spread out the posterior distribution is; it does not by itself prove that the probabilities are calibrated.

VI. Results and Discussion

A. Bayesian Network Results

The BN correctly classifies 333 of the 440 holdout samples, corresponding to 75.68% accuracy. Its macro F1-score is 0.73 and weighted F1-score is 0.73. These results show that the current discretized BN captures useful patterns, but the class-wise report also indicates uneven performance across crop labels.

B. Comparison with Conventional Classifiers

Table II presents the holdout results from the same train-test partition. Random Forest, Gaussian Naive Bayes, and Extra Trees achieve the highest accuracy at 99.55%, considerably higher than the current BN result.

The results show that the BN is not the most accurate classifier on this dataset. Its probabilistic structure supports posterior inference and scenario analysis, but the current experiment does not establish that these properties improve decision quality. Probability calibration and proper scoring rules were not evaluated for the BN because its full holdout posterior vectors were not retained.

Figure 1 presents the class-wise F1 scores of the Bayesian Network. The model performs strongly for several crop classes, while lower scores for blackgram, maize, and kidneybeans indicate that performance is uneven across the 22 classes. This class-wise view complements the aggregate accuracy and motivates further investigation of discretization and network structure.

The confusion matrix in Fig. 2 shows the crop classes that the Bayesian Network predicts correctly and those it confuses. The diagonal represents correct predictions, while off-diagonal entries identify class-specific errors.

TABLE II
HOLDOUT COMPARISON ON THE SAME STRATIFIED 80:20 SPLIT

Model	Accuracy (%)	Macro F1	Weighted F1
Random Forest	99.55	0.9955	0.9955
Gaussian Naive Bayes	99.55	0.9954	0.9954
Extra Trees	99.55	0.9955	0.9955
Gradient Boosting	98.86	0.9887	0.9887
RBF SVM	98.41	0.9840	0.9840
Decision Tree	97.95	0.9794	0.9794
KNN	97.95	0.9793	0.9793
Logistic Regression	97.27	0.9725	0.9725
Bayesian Network	75.68	0.7300	0.7300

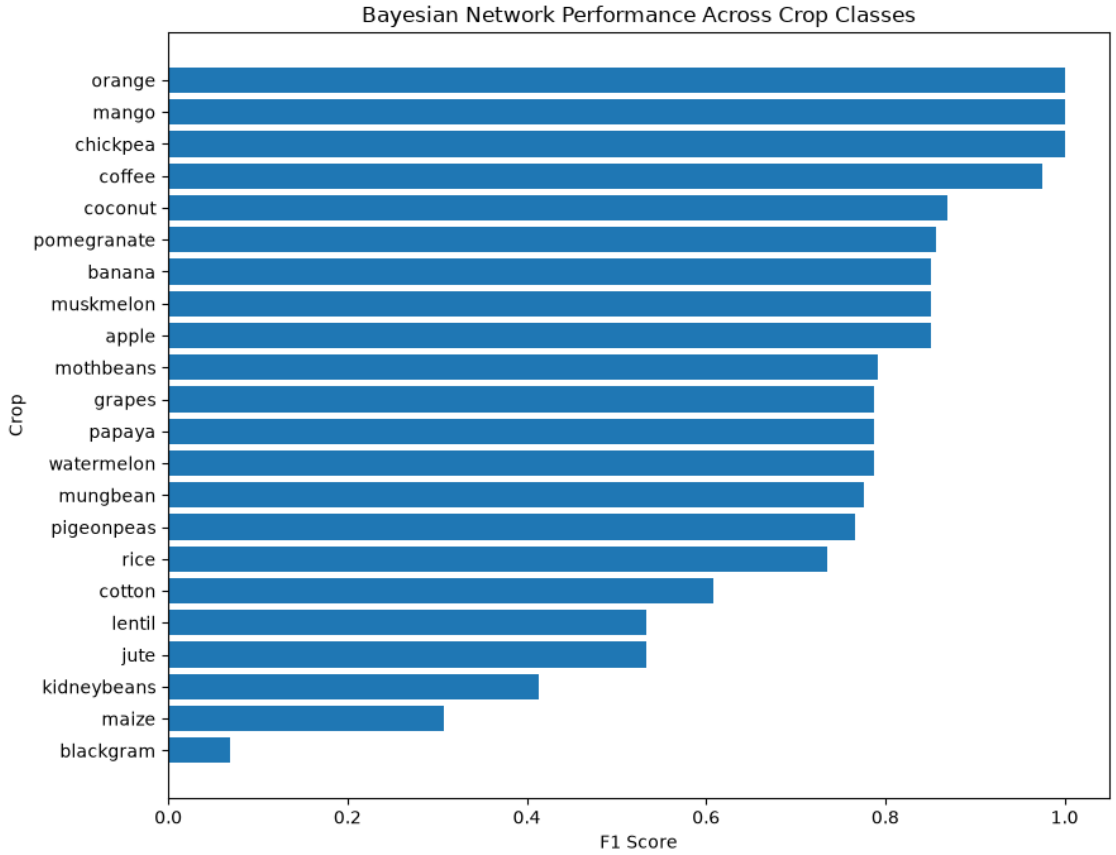


Fig. 1. Class-wise F1 scores of the Bayesian Network crop classifier.

C. Five-Fold Cross-Validation

Five-fold stratified cross-validation is performed only on the training partition for conventional baselines, using shuffled stratified folds with a fixed random seed. Table III reports mean accuracy and sample standard deviation across the five folds. These values were recomputed from the supplied CSV using the model definitions described above.

The conventional baselines retain high scores across the training partition's folds. The BN has not yet been evaluated using an equivalent repeated cross-validation procedure; therefore, a direct cross-validation comparison with the BN remains future work.

D. Rainfall Scenario Analysis

For the selected baseline evidence, the most probable crop changes when rainfall evidence changes. Table IV reports the top crop and normalized entropy for each scenario.

In this selected scenario, High rainfall evidence produces a more concentrated posterior distribution than Medium rainfall evidence. This result describes the fitted model's response to the chosen evidence and should not be interpreted as a general agricultural rule. The baseline evidence is categorical and represents a controlled model experiment rather than a field trial.

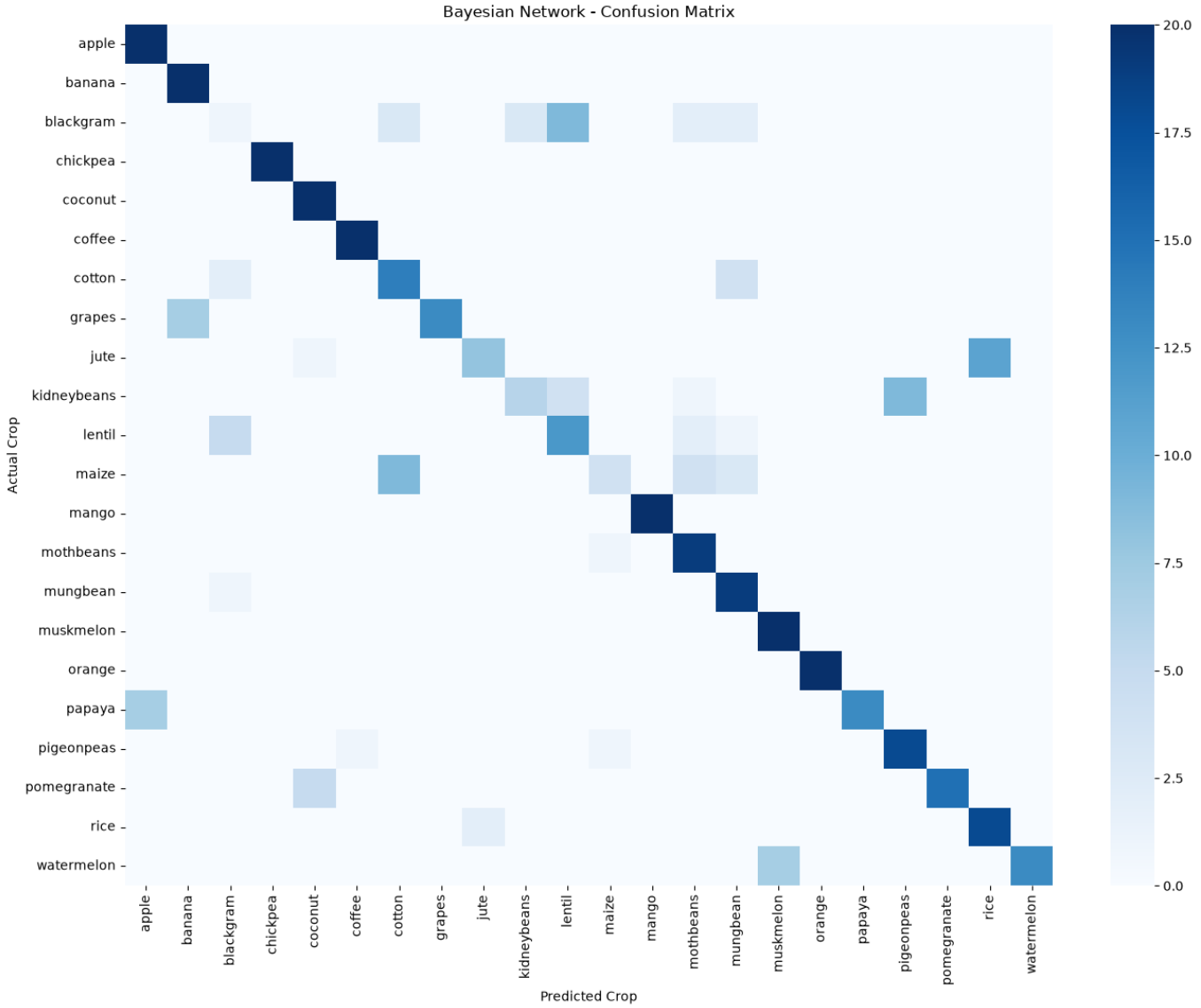


Fig. 2. Confusion matrix of the Bayesian Network on the 440-sample holdout test set. Rows represent actual crop classes and columns represent predicted classes.

TABLE III
FIVE-FOLD CROSS-VALIDATION ON THE TRAINING PARTITION

Model	Accuracy (%)	Macro F1
Gaussian Naive Bayes	99.49 ± 0.47	0.9949 ± 0.0047
Random Forest	99.38 ± 0.68	0.9937 ± 0.0069
Extra Trees	99.38 ± 0.37	0.9937 ± 0.0037
Gradient Boosting	98.69 ± 0.38	0.9869 ± 0.0040
Decision Tree	98.52 ± 0.76	0.9852 ± 0.0076
RBF SVM	97.90 ± 1.15	0.9788 ± 0.0119
Logistic Regression	96.82 ± 0.74	0.9678 ± 0.0076
KNN	96.53 ± 1.35	0.9650 ± 0.0139

TABLE IV
BN RESULTS FOR THE SELECTED RAINFALL SCENARIOS

Rainfall	Top Crop	Probability (%)	Norm. Entropy
Low	Blackgram	33.61	0.5858
Medium	Kidneybeans	43.06	0.6682
High	Pigeonpeas	66.39	0.4407

E. Feature Sensitivity

Table V reports the maximum total variation distance observed when each feature is varied over Low, Medium, and High while other evidence remains fixed.

Phosphorus and potassium produce the largest changes in the posterior distribution under the selected evidence. Nitrogen, temperature, and pH produce no change in this scenario. A zero change should not be interpreted as evidence that these variables are unimportant in agriculture: it may arise from the learned network structure, conditional probability tables, discretization,

TABLE V
FEATURE SENSITIVITY FOR THE SELECTED BASELINE EVIDENCE

Feature	Maximum TV Distance
Phosphorus	0.9578
Potassium	0.9459
Humidity	0.5997
Rainfall	0.5112
Nitrogen	0.0000
Temperature	0.0000
Soil pH	0.0000

or the selected evidence configuration. Further analysis over multiple baselines and alternative learned structures is needed.

F. Limitations

The current BN uses a single stratified holdout split and a three-bin discretization scheme. The results may depend on the chosen quantile boundaries, scoring function, prior pseudo-counts, and learned network structure. The benchmark dataset is balanced and relatively small; its provenance and real-world representativeness must be verified. The baseline classifiers have not been tuned through a dedicated hyperparameter search, and their high results should not be generalized to other datasets without validation. The current BN evaluation also lacks exported full posterior vectors for the holdout set, preventing a direct Brier-score, log-loss, and calibration comparison with conventional models. The confusion matrix shows the distribution of errors, but individual cell counts are not annotated in the supplied figure; readers should therefore use it to inspect broad error patterns rather than infer exact counts from color intensity. The unusually high baseline scores also warrant a reproducibility check of feature columns, label handling, and the train-test split before any broader claims are made.

VII. CONCLUSION

This paper evaluated a discretized Bayesian Network for crop recommendation using seven soil and environmental features and 22 crop classes. On a stratified 80:20 holdout split, the BN achieved 75.68% accuracy and a macro F1-score of 0.73. Conventional classifiers achieved substantially higher accuracy on the same split, with Random Forest, Extra Trees, and Gaussian Naive Bayes reaching 99.55%. The BN's scenario analysis showed that posterior crop probabilities and normalized entropy vary with rainfall evidence. However, the observed sensitivity is conditional on the selected baseline evidence and does not establish causal agricultural relationships. The present results therefore support the BN as a probabilistic baseline, not as a more accurate alternative to conventional classifiers. Overall, the project successfully implemented and evaluated a Bayesian Network-based crop recommendation system and examined its performance from both classification and probabilistic perspectives. Although conventional classifiers provided better predictive accuracy on the evaluated dataset, the Bayesian Network offered an explicit probabilistic representation for posterior inference, environmental scenario analysis, and uncertainty assessment. The experiments provide a transparent account of the model's capabilities and limitations on the selected dataset.

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